NETWORKS AND COMPLEXITY

Solution 17-1

This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at https://github.com/NC-Book/NCB

Ex 17.1: Simple timescale renormalization [**Lvl. 1**]

Change the timescale to make the parameter k disappear from the system

$$\dot{x} = k(x - x^2)$$

Solution

We define

$$\tau = at. \tag{1}$$

Hence

$$\frac{\partial t}{\partial \tau} = \frac{1}{a}. \tag{2}$$

Therefore

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \frac{\partial t}{\partial \tau}\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{a}k(x - x^2) \tag{3}$$

Now τ is our new time, and we define

$$\dot{x} = \frac{\mathrm{d}x}{\mathrm{d}\tau} \tag{4}$$

which allows us to write

$$\dot{x} = \frac{k}{a}(x - x^2). \tag{5}$$

Choosing $a = k$ yields the desired equation

$$\dot{x} = (x - x^2) \tag{6}$$

Note we can always set one of the rates in the system to one by means of a timescale renormalization. This just means we measure all other rates relative to the rate we set to one. So for example if you have two processes which have rate constants $a = 10/\mathrm{s}$ and $b = 20/\mathrm{s}$, then you can renormalize time such that $a = 1$ and $b = 2$ (without the unit seconds). This makes the a disappear and the rate b is now measured in multiples of a instead of seconds.